Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Tough structural thermosets with microstructure by additive manufacturing			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification 1)	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Chemicals			
	Medium Category	Small Category	Medium Category	Small Category
	Polymeric materials	Polymer processing technologies		
Concept of Proposed Technology	The proposed project will develop a technology that combines photopolymerization-induced phase separation (photo-PIPS) and 3D printing to create thermosets with multiscale microstructure that conveys enhanced toughness, strength and stiffness. By "microstructure" here, we refer to subdomains of a soft and ductile phase alternating with harder and more brittle subdomains. This new technology may be used to control toughness and strength spatially and to incorporate functionally graded regions that mitigate stress concentration to prevent fracture and debonding in multicomponent/multimaterial systems. The state of the art in toughening thermosets is based on the incorporation of rubber microparticles or nanoparticles, which is subject to particle mixing challenges and does not allow for detailed spatial control of additive distribution (1,2). Our technology uses a bio-inspired approach: bone and mollusk shells/nacre have high strength and stiffness and inherit their remarkable toughness from the microstructure containing nano- and micro-			

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	scale soft/tough subdomains interspaced with strong/stiff subdomains (3,4). We aim to produce thermosets with spatially controlled microstructure, strength and toughness designed for specific applications. Areas of potential applications include coatings, wearable soft materials embedding hard components (e.g., electronics), structural components of complex geometry, components that are subjected to environmental embrittlement, etc. The proposed technology is based on our preliminary work published in (5,6,7,8).
Rationale for Joint R&D Project	⬭ Mitigating the intrinsic brittleness of thermosets. Thermosets are widely used for their high stiffness and strength and their stability, but their inherent toughness, impact resistance, and scratch resistance in the absence of additives are low. While some commercial solutions have been developed over the past decades (1,2), these are not applicable for general situations. Our proposed solution is applicable in the context of additive manufacturing (UV-cured thermosets) and hence offers high manufacturing flexibility. ⬭ Coupling microstructure and component design for optimal mechanical performance. The proposed technology creates the desired microstructure (by phase separation) on scales smaller than the resolution of 3D printing, leading to products with truly multiscale structure. The local material properties may be modulated based on the component design: re-entrant corner stress concentration may be mitigated, probability of debonding and fracture at stiff embeddings (e.g. electronic components embedded in soft wearables) is reduced. ⬭Collaboration between RPI and Korean SMEs may evolve into a mutually beneficial synergetic partnership.
Key Words	1. Thermosets 2. Photo-polymerization 3. Phase separation 4. Toughness and mechanical properties 5. Coupled component and microstructural design

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	⬭Objective 1: Demonstrate the ability to produce phase separation within photo-polymerizable thermosets (acrylates) using a variety of phase separating additive polymers in UV-cured castings. ⬭Objective 2: Demonstrate enhanced toughness controlled by the microstructure in cast samples and samples printed with spatially nominally uniform printing parameters. ⬭ Objective 3: Demonstrate microstructure development on scales smaller than the scale of printing, print with spatially non-uniform printing parameters to show control of the multiscale structure of the material. ⬭Objective 4: Develop methods to couple component design with microstructural design to provide optimal mechanical performance in specific application(s) (e.g. protection against fracture and debonding at stress concentrators). ⬭Objective 5: Demonstrate these advances in the context of specific applications of SME.			
R&D Target Specifications	⬭Control material property modulation (contrast in stiffness and ductility) on scales between 100 nm and 100 microns by phase separation and on scales larger than 100 microns by 3D printing (DLP). ⬭Achieve toughness $K_{Ic} > 2.5\ MPa\sqrt{m}$, (compare with typical acrylate toughness of < $1\ MPa\sqrt{m}$), with strength larger than 50 MPa. ⬭ Develop methods to couple component design and microstructural design to eliminate stress concentration and reduce failure probability. Demonstrate the technology to produce such materials.			
R&D Plan	The proposed work has two thrusts: Thrust 1: Material and manufacturing method development, and Thrust 2: Coupled component and microstructural design. The concept underlying Thrust 1 is shown in Figure 1. Parts will be produced by digital light processing (DLP) in which digital masks will be used to modulate exposure to UV radiation (9). This allows defining patterns with the printing resolution of about 50 to 100 microns. Phase separation takes place during photo-polymerization and leads to the formation of sub-domains composed predominantly of a polymeric additive added to the monomeric bath for this purpose. The number density and size of these sub-domains may be adjusted by controlling the kinetics of polymerization (via the UV light intensity or intermittent light exposure) and the nature and chain length of the phase separating additive (6,7). The resulting microstructure has multiple scales: phase separation produces features smaller than the printing resolution, down to ~100 nm in size, while patterning on scales larger than the printing resolution can be achieved DLP.			

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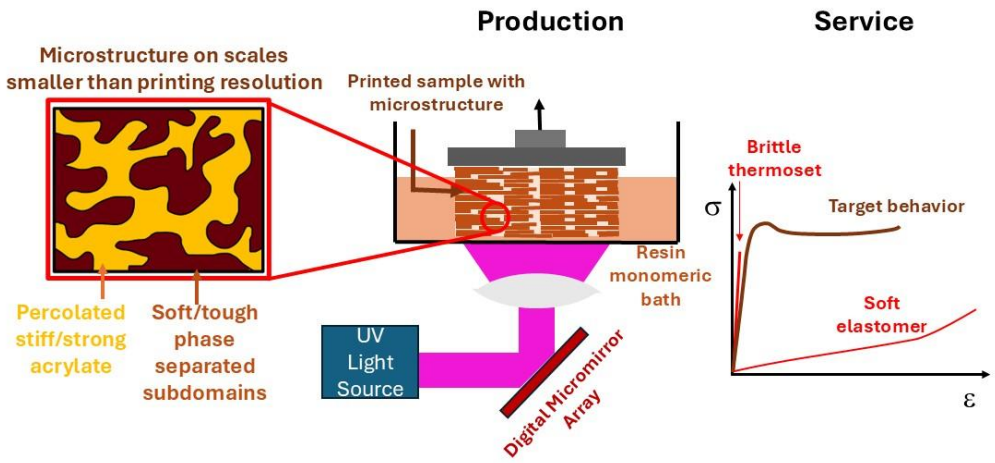

Figure 1. Schematic of the proposed concept.

The contrast in mechanical properties between the two types of sub-domains is critical for achieving the target performance. This will be controlled by the chemical composition of the monomeric bath and additives. Combinations of mono-functional, di-functional and tri-functional acrylates will be used to construct the molecular network, while linear polymers will be used as phase separating additives. The technology includes two scenarios: the phase separating phase may be a thermoplastic or a thermoset which is chemically and structurally different from the acrylate molecular network forming the base phase. In both cases, the phase separating phase will be more ductile and tougher than the base phase, therefore allowing for crack trapping and enhanced toughness (4).

The size of the subdomains will be controlled by the nature of the phase separating additives: longer chain additives phase separate more effectively leading to larger subdomains (6,7,8). Note that the chemistry is such that no phase separation takes place in the monomeric bath, before UV-driven polymerization. An additional method to control phase separation is to adjust the rate of photopolymerization. For example, acrylate chain polymerization is faster than methacrylate polymerization (10). When these photo-polymerize together, some degree of phase separation of the two species takes place, even in the absence of phase separating moieties (5). Additive control and kinetics control will be used to adjust phase separation subdomain size.

In Thrust 2 we will develop methods and computational tools to couple component design to the design of the microstructure. The objective is to devise a unitary framework that provides process (printing) parameters based exclusively on the desired component shape and type of loading in service, Fig. 2. Note that process parameters are supposed to be variable in time during printing to allow the development of spatially varying microstructure.

Defining process parameters requires:

- Task (2a): design the spatial distribution of material properties (stiffness and toughness), i.e. functional gradients, that mitigate stress concentration and potential microdamage that may occur in service.
- Task (2b): based on (2a), define the microstructure (volume fraction of soft/tough sub-domains, size of sub-domains) and its spatial distribution.

- Task (2c): identify printing parameters (exposure time and UV light intensity) that produce the desired microstructure.

Task (2a): start from the stress field in the component of interest subjected to service loads and adjust the spatial distribution of material properties to render the stress field as uniform as possible. This computational tool will be developed in this project.

Tasks (2b) and (2c): a large number of process conditions will be used to produce samples with nominally homogeneous structure. These samples will be fully characterized microstructurally and mechanically. This will create a database relating process parameters, structural information and mechanical performance.

A machine learning-driven (ML) model will be trained based on this information and will be subsequently used to predict process parameters for target properties. The ML model will be then used to define the time variation of process parameters during printing to produce heterogeneous components with material properties distribution as defined in Task (2a).

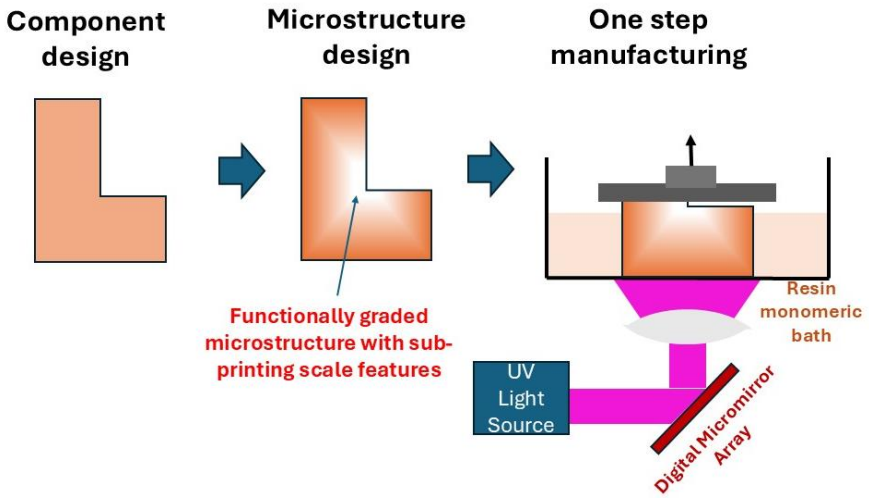

Figure 2. Coupled component and microstructural design define process/printing parameters.

Milestones and timeline:

Phase 1 (6 months): In phase 1 we will reach Objective 1, i.e. will produce polymerization-induced phase separation within photo-polymerizable thermosets (acrylates) using a variety of phase separating additive polymers. We will also demonstrate the feasibility of Objective 2 - enhanced toughness relative to the base acrylate.

Phase 2:

12 months: We will accomplish Objective 2, i.e. will demonstrate enhanced toughness controlled by the microstructure in cast samples and samples printed with spatially uniform printing parameters. For this purpose, we will down-select one or a small set of compositions adequate for the application(s) discussed with the SME.

18 months: We will perform printing with spatially non-uniform parameters to show control of the

	multiscale structure of the material. We will also begin developing the database relating printing conditions, structural parameters and material properties. 24 months: A broad range of material combinations will be explored to complete the process-structure-properties database. We will begin developing computational tools that predict the distribution of material parameters for target (uniform) stress distribution in samples subjected to given service loads, Objective 4. 30 months: The database will be used to train and validate the ML tool. ML predictor will be used in conjunction with the computational tool developed in the previous quarter. Application to specific parts of interest for the SME. 36 months: We will finalize Objective 4 and will work with the SME to achieve Objective 5, technology transfer.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	⬭Rensselaer Polytechnic Institute (RPI) will perform all research addressing Objectives 1 to 4. The specific tasks and steps to be taken are described under Milestones and timeline in section 3. ⬭RPI will consider a broad range of acrylate monomers and molecular additives. However, interaction with SME will be critical in narrowing down this range to address the needs of the application. ⬭RPI will develop computational and ML tools that control the printing process for target spatial distribution of microstructure. ⬭ Demonstrate the advance on components of interest for the SME.
Expected R&R of Korean SME Partner	⬭ The SME will interact with RPI in defining the application and target performance. ⬭ The SME will provide information about specific polymeric components they seek to incorporate in the thermosets to be developed (e.g. for providing additional functionality required by the application). If such components are not available commercially, the SME will provide RPI with these materials. If no such specific requirements are made, RPI will develop the material based on existing data and technology. ⬭ RPI and the SME will work together on adapting the computational and ML tools to be developed to the needs of the company, Objective 5.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The outcomes of the proposed research are: 1) UV-curable thermosets with multiscale microstructure having enhanced toughness and strength, reaching or surpassing the specifications outlined in section 3. 2) The technology to produce such materials using a combination of photo-polymerization-induced phase separation and 3D printing (DLP). 3) Computational and ML-driven methods allowing the control of the printing process to achieve functionally graded microstructures, which provide spatially varying material properties that mitigate damage from stress concentrators and improve component life. This set of outcomes defines a complete strategy to produce parts of specified geometry and in-service performance starting from computational design and including process control and manufacturing. This technology applies to a variety of cases, including bulk parts as well as films, coatings, and interfaces. The method used to predict printing parameters function of the in-service component performance is flexible, transferable, and applicable to arbitrary geometries. The impact is related to (i) the improvement of the toughness of UV-sensitive thermosets used for bulk or protective coatings, and (ii) advancement of manufacturing technology by coupling component and microstructural design, and integration with component processing. This concept allows extensions: microstructural design and development via photo-polymerization-induced phase separation for functionalities beyond toughening. RPI will work with the SME on the implementation of this technology in the context of applications of interest for the Korean partner.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization